INORGANIC CUMULATIVE EXAMINATION

April 28, 1990

The following are formulas for structure drawings (1-10), figure captions (11-15), or table headings (16-18) from a chapter entitled "d- and f-block organometallics" in the new inorganic text by Shriver. Each structure drawing, figure, or table is obviously included to illustrate a special point or be an example of an important general rule. Choose 10 from the list and give the likely reason for the inclusion of that drawing/figure/table in the text. This exam will be graded according to the level of sophistication of your answers.

1. $[PtCl_3(C_2H_4)]^-$
2. $Fe(C_5H_5)_2$
3. $Ti(\eta^1\text{-}C_5H_5)_2(\eta^5\text{-}C_5H_5)_2$
4. $W(\eta^3\text{-}C_5H_5)(\eta^5\text{-}C_5H_5)(CO)_2$
5. *trans*-$[IrCl(CO)(PPh_3)_2]$
6. $Ni(cod)_2$
7. $Co(\eta^3\text{-}C_3H_5)(CO)_3$
8. $Ru(\eta^4\text{-}C_8H_8)(CO)_3$
9. $[Ni_2(C_5H_5)_3]^+$
10. $Co_4(CO)_{12}$
11. The influence of metal-ligand π bonding on CO bonds in metal carbonyls.
12. The IR spectrum of $(C_5H_5)_2Fe_2(CO)_4$.
13. Observed ^{1}H-NMR spectra for $(\eta^4\text{-}C_8H_8)Ru(CO)_3$ at various temperatures.
14. Representative cluster compounds.
15. σ, π, and δ interactions between two *d*-block metal atoms situated along the *z*-axis.
16. The influence of coordination and charge on CO stretching wavenumbers.
17. Tolman cone angle for various ligands.
18. Examples of metal-metal bonded tetragonal prismatic complexes.